Shreya Prakash

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Research Areas: Causal Inference, Experimentation, Machine Learning

Website: https://shreyap18.github.io/

EDUCATION

University of Washington June 2026

PhD in Statistics, Advanced Data Science Track *(Advisor: Elena Erosheva)*

Thesis: "Statistical Advancements in Causal Decomposition and Causal Discovery for Real-World Applications"

Coursework: Causal Inference, Statistical Learning, Advanced Statistical Theory, Bayesian Statistics, Advanced Regression, Fairness in Machine Learning, Computational Biology, Measure Theory

Carnegie Mellon University December 2019

B.S in Statistics and Machine Learning, University Honors

Coursework: Machine Learning, Statistical Graphics and Visualization, Parallel and Sequential Data Structures and Algorithms (C/SML), Linear Algebra

WORK EXPERIENCE

Applied Science Intern, *Amazon* Summer 2024, 2025

- Developed novel Bayesian methodology to reduce noise in feature-impact estimates for A/B tests in Weblab, Amazon's internal experimentation platform supporting 100K+ large-scale experiments annually. My internship work is currently in production.
- Developed instrumental-variable survival models to study causal factors of workforce attrition as part of Amazon's Execution Planning Science (EPS) team, which develops science solutions for labor planning and operations optimization.

Data Scientist*, Marinus Analytics* February - September 2020, Summer 2021

- Applied machine learning and time series analysis for unstructured child welfare case records.
- Launched spam filter and underage person detection algorithms for TraffickJam: an application that uses human trafficking advertisement data to aid law enforcement with finding trafficking victims and traffickers.
- Productionalized Infoshield: a text clustering algorithm for large scale human trafficking advertisement datasets.

PUBLICATIONS

- **Prakash, S.**, Xia, F., & Erosheva, E. (2026). *Causal discovery via statistical power (CDSP).* **In review**. arXiv preprint arXiv:2605.13550. https://arxiv.org/abs/2605.13550.
- **Prakash, S.**, Xia, F., & Erosheva, E. (2026). *Hypothesis testing for functional causal discovery.* **In review.**
- **Prakash, S.**, Erosheva, E., & Lee, C. (2026). *Causal decomposition of Black-white disparities in NIH's selection of proposals for panel discussion.* **In review.** Student Paper Competition Winner (ASA SSS/GSS/SRMS 2026).
- Samir, F., Ahn, E., **Prakash, S.**, Soskuthy, M., Shwartz, V., & Zhu, J. (2025). *Efficiently identifying low-quality language subsets in multilingual datasets: A case study on a large-scale multilingual audio dataset.* Transactions of the Association for Computational Linguistics (**TACL**). https://doi.org/10.1162/tacl_a_00759.
- **Prakash, S.**, Xia, F., & Erosheva, E. (2024). *A diagnostic tool for functional causal discovery.* arXiv preprint arXiv:2406.07787. https://arxiv.org/abs/2406.07787. **Software:** causalDiagnose R package. Presented at the American Causal Inference Conference (2024) and the Western North American Region of the International Biometric Society (2023).

SKILLS & AWARDS

Programming: Proficient in R, Python, SQL. Familiar with C, Standard ML, Matlab, Java, Bash.

Libraries/Software: numpy, pandas, scipy, sklearn, statsmodels, tensorflow, seaborn, matplotlib, rjags, tidyverse, parallel, Git, Docker, AWS S3 & EC2, Keras.

Awards: ASA SSS/GSS/SRMS Student Paper Competition Winner (2026); Rising Star Award, Oden Institute for Computational Engineering and Sciences (2025); CSSS Travel Award, University of Washington (2023).